DESCRIPTION OF COURSEWORK

Course Code	EEE308
Course Name	RF&Microwave Circuits
Lecturer	Associate Prof. Dr. Zulkifly Abbas
Academic Session	2025/04
Assessment Title	Lab: Development of Specification Data Sheet for ATM PNR- C122D-10 Directional Coupler

A. Introduction/ Situation/ Background Information

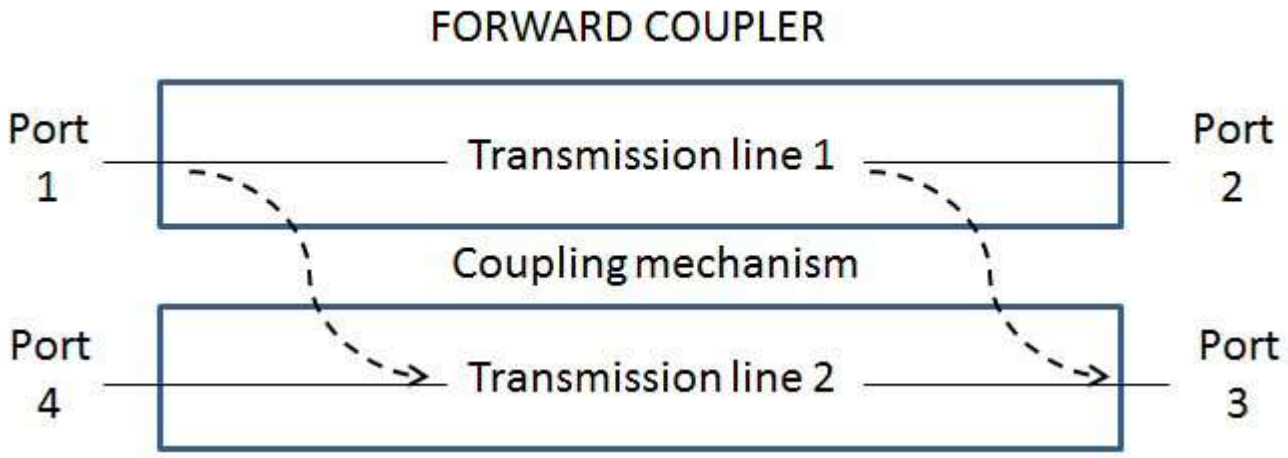
Directional couplers are four-port circuits where one port is isolated from the input port. Directional couplers are passive reciprocal networks. All four ports are (ideally) matched, and the circuit is (ideally) lossless. Directional couplers can be realized in microstrip, stripline, coax and waveguide. They are used for sampling a signal, sometimes both the incident and reflected waves (this application is called a reflectometer).

Directional couplers generally use the distributed properties of microwave circuits. The coupling process generally occurs within a quarter-wavelength or multiple quarter-wavelength portions of the device. In these distributed couplers the energy and fields of one structure interact with the energy and fields of another distributed structure, coupling signals from one to another. Lumped element and hybrid couplers can be constructed as well. These couplers do not rely on the interaction of the fields and waves between transmission structures to create performance similar to that of coupled line structures but instead rely on networks composed of independent (un-coupled) circuit elements.

A directional coupler has four ports, where one is regarded as the input, one is regarded as the "through" port (where most of the incident signal exits), one is regarded as the "coupled" port (where a fixed fraction of the input signal appears, usually expressed in dB), and one is regarded as the "isolated" port, which is usually terminated. On some commercial couplers the fourth (isolated) port is terminated internally or externally with a matched load - making a four port device seem like a three port device. If the signal is reversed so that it enter the "through" port, most of it exits the "input" port, but the coupled port is now the port that was previously regarded as the "isolated port". The coupled port is a function of which port is the incident port.

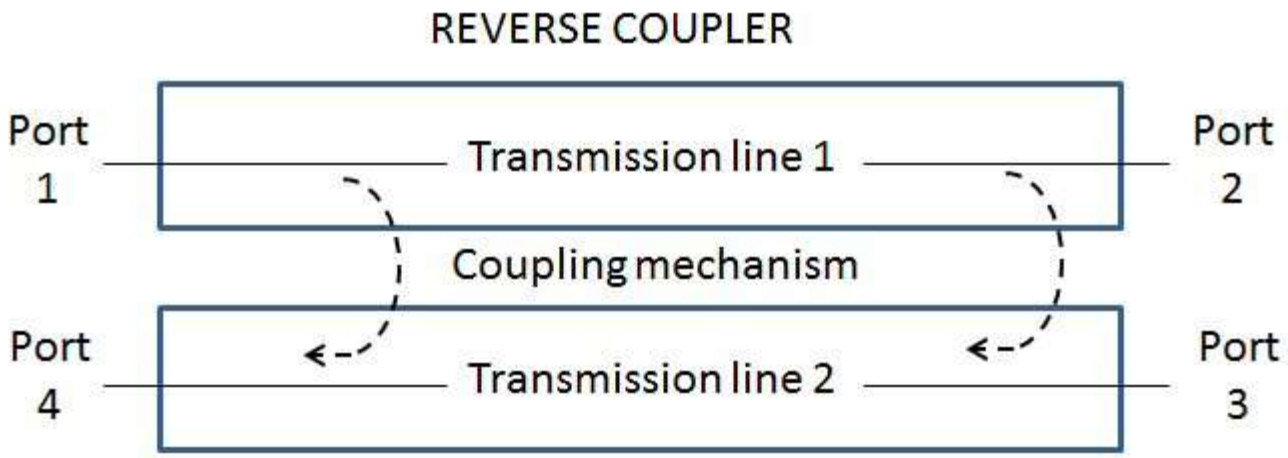
Forward-wave versus backward-wave couplers

In the *forward-wave* coupler, energy that propagates down transmission line starts a parallel wave down transmission line 2 as shown below. The most common forward coupler is the multi-hole coupler realized in waveguide. In this case the holes are spaced a quarter wave apart so that the reverse wave cancels out. Flat coupling across a wide bandwidth is possible, using a lot of math to specify the sizes and positions of each coupling hole. For a tutorial on coaxial directional coupler, refer to
https://directional-coupler.com/tutorials/functional-principle-of-a-directional-coupler/



In the *backward,* or *reverse* coupler, energy that propagates down transmission line starts a reverse wave down transmission line 2, as shown below. Single-section coupled transmission lines are always backward-wave couplers (and outputs are in quadrature), the Lange coupler is one example. Asymmetric multi-section coupled structures provide forward-wave, in-phase response, while symmetric multi-section couplers provide backward-wave, quadrature response.

https://directional-coupler.com/tutorials/functional-principle-of-a-directional-coupler/



There is no "standard" way to number the ports. We have used a clockwise notation for consistency. Also, there is no perfect coupler, in a forward coupler there will be reverse coupling to some degree, and vice-versa.

Most waveguide couplers couple in the forward direction as they rely on multiple coupling holes; a signal incident on port 1 will couple to port 3 (port 4 is isolated in our clockwise notation). Microstrip or stripline couplers are backward wave couplers because they rely on coupled lines: for a signal incident on port 1, port 4 is the coupled port (port 3 is isolated in our clockwise notation).

The coupled port on a microstrip or stripline directional coupler is closest to the input port because it is a *backward wave coupler*. On a waveguide broadwall directional coupler, the coupled port is closest to the output port because it is a *forward wave coupler.*

Forward couplers are in-phase couplers. Backward couplers couple in quadrature (the coupled port phase is 90 degrees more negative than the direct port).

The ATM coupler below is a coaxial type according to the manufacturer (you have to cut it apart to know that), which means it is a backward wave coupler. The input port is printed on the right, and the port facing up is the coupled port(the opposite port is terminated with that weird cone-shaped thingy which voids the warrantee if you remove it.



Directional Coupler Testing Process by Fieldfox(Vector Network Analyzer)

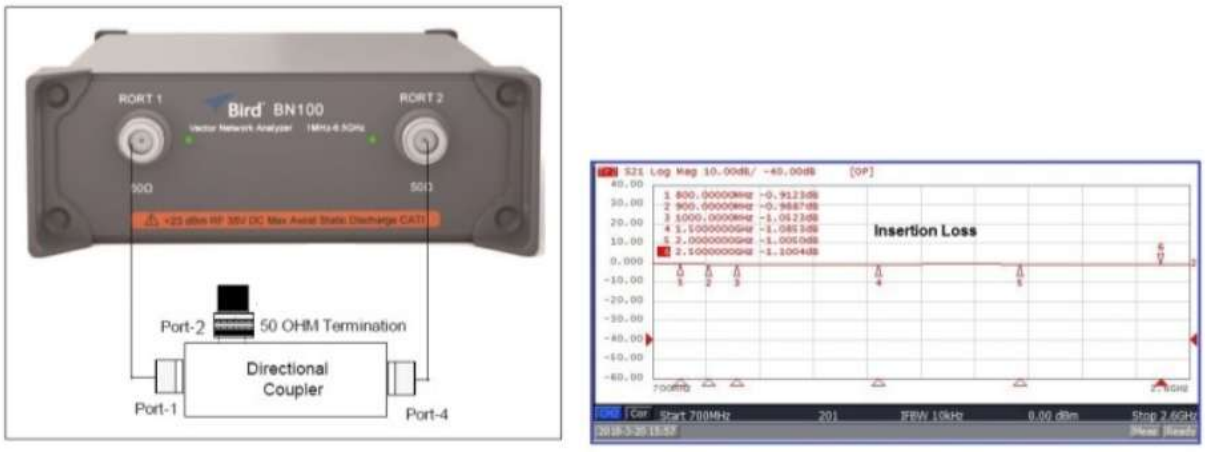
1.Coupling Power :
The ratio of the power that the coupled output port is sampled from the input port when all ports are terminated with reflection less terminations. The nominal coupling is specified as the average of the maximum and the minimum coupling within the frequency band.



(Illustration Purpose Only. Bird BN100 applicable to Keysight Field Fox Analyzer)

Connect the coupler to the Network Analyzer as indicated in figure above. Note the output port is terminated with a 50 W load. The measurement values can be read at S_{21} in Field Fox.

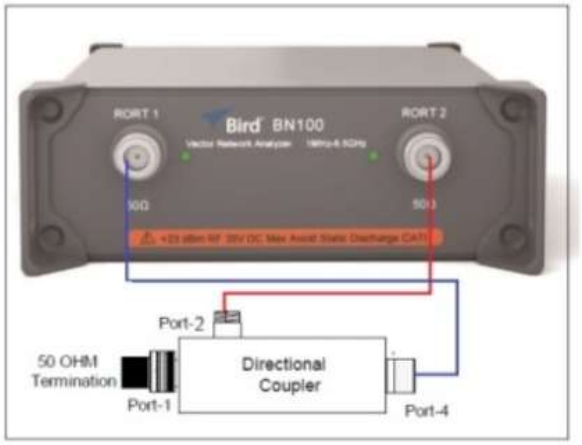
2. Insertion Loss : The net unrecoverable power in dB dissipated within the circuit at any frequency within the specified range.



Connect the coupler to the Network Analyzer as indicated in figure above. with the coupled port terminated with a 50 W load . The measurement values can be read at S_{21}.

Directivity : requires two measurements at Port 4: one with Port 2 terminated in a perfect termination and one with a perfect short.
For example, the reflection port (Port 4) of a directional coupler measures -20 dBm when Port 2 is terminated with a short and -55 dBm with a perfect termination, the directivity is simply the difference of the two, 35 dB.

Isolation : Isolation is measured between the output port and coupled port of the directional coupler to ensure the quality of internal circuit and design perfection.





Connect the coupler to the Network Analyzer as indicated in figure above. with the input port terminated with a 50 W load . The measurement values can be read at S_{21}.

Assignment for this coursework: Specification Data Sheet for ATM PNR- c122D-10 Directional coupler

A specification data sheet, or simply a data sheet, is a document that summarizes the technical characteristics and performance of a product, machine, component, material, or software. It is used by buyers and engineers to understand what the product is, how it should be used, and how it compares to other products.

Submit a Lab report with detailed measurement results and analysis of ATM PNR C122D-10 Directional coupler. Summarize your results in a Specification Data Sheet format using the following reference as a guideline: *Mini-Circuits' ZCDC20-E18653+ Datasheet, 2012*

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 3 Build microwave circuits and measure its signals

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
   i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).
   ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.
   iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.

2. All assessments submitted must be the student's own work, without any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. Any form of academic misconduct, including using prohibited materials or inappropriate assistance, is a serious offense and will result in a zero mark for the entire assessment or part of it. If there is more than one guilty party, such as in case of collusion, all parties involved will receive the same penalty.

D. Instruction to Students

The report must be submitted (softcopy/ hardcopy) before end of week 13. The reports must include abstract ,introduction, theory, methodology, results &discussion, conclusion and references.

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	Lab Report– Development of Specification Data Sheet for ATM PNR-C122D-10 Directional Coupler	100%
	TOTAL	**100%**

F. Task(s)

Submit a Lab report with detailed measurement results and analysis of ATM PNR C122D-10 Directional coupler using N9932A Field Fox Analyzer. Summarize your results in a Specification Data Sheet format using the following reference as a guideline: *Mini-Circuits' ZCDC20-E18653+ Datasheet, 2012*

Lab Report must have the components outlined in the rubrics in Appendix 1 and submitted before end of week 13.

APPENDIX 1

MARKING RUBRICS

	Excellent (90-100%)	**Good (80-89%)**	**Adequate (60-79%)**	**Poor (20-60%)**	**Lacking 20-0%**
Abstract/ Introduction/ Theory	1.Includes complete well-written information on the background to understand the subject of the experiment and its significance. 2.Summarizes what is already known on the subject. 3. Includes the question to be answered by the lab and states hypothesis that is based on research and/or sound reasoning, described in the background. 4. Includes the description on how the hypothesis is going to be tested. 5. Deep understanding of the subject as well as creativity is demonstrated.	1.Includes strong introductory information, most major points provided. 2.Most important details in the summary provided. 3. Most important details of the hypothesis covered. 4. An idea on how to test the hypothesis is included. 5. Good understanding is demonstrated.	1.Includes good introductory information, missing some points in the summary. 2.Hypothesis is missing some minor details. 3.No description on how the hypothesis is going to be tested is provided. 4. Appropriate understanding of the subject is demonstrated.	1. Introductory information provided, missing some major points. 2. Important details of the hypothesis are missing. 3. No description on how the hypothesis is going to be tested provided. 4. Limited understanding of the subject is demonstrated.	1. Part is missing or plagiarized. OR 2.Little background information is present. 3.Incorrect information is provided. 4. For the hypothesis, important experimental details are missing, of the hypothesis is missing. 5. Poor understanding or no understanding of the subject is demonstrated.
methodology	Description (not a copy of a protocol!) is included, detailed enough so it could be repeated by another scientist. Description is not overloaded with unnecessary details.	Description included, some steps are vague or unclear OR instead of description, a step-by–step protocol is	The description gives generalities, enough for reader to understand how the experiment	Would be impossible to repeat the experiment, reader must guess at how the data was gathered or	Part is missing or plagiarized

		copied from manual.	was conducted	experiment conducted	
Results	Results and data are clearly recorded, so it is easy for the reader to see trends. Excellent graphs, tables or figures with appropriate names and labels, excellent information and understanding of results, excellent interpretation of data and trends while referring to figures and tables.	Results are clear, but trends are not obvious or there are minor errors in organization. Good figures, graphs, or tables, good information results, good understanding of results, but some figure/table names or labels are missing.	Results are unclear, trends are not obvious, disorganized, but there is enough data to show the experiment was conducted. Appropriate figures, graphs or tables or drawings, some labels and figure names are missing. Only some understanding of results is evident.	Results are disorganized or poorly recorded, do not make sense; not enough data was taken to justify results. Missing or incorrect information.	Part is missing or plagiarized. Misleading, confusing and/or incorrect information. Poorly presented results.

Discussion/ Conclusions	1.Summarizes data used to draw conclusions 2.Conclusions follow data (not wild guesses or leaps of logic) 3. Hypothesis is rejected or accepted based on the data. 4. Discussion of applications or realworld connections is present. 5. Deep understanding of the subject as well as creativity is demonstrated.	Good conclusion, important information provided, or one point from Excellent is missing	Appropriate conclusion. Two points from Excellent are missing.	Limited conclusion. Important information is missing.	Part is missing or plagiarized. OR No conclusion is made. Missing important information. Little effort or reflection is shown.
References	All references and citations are correctly written and present. More than two resources are cited. Student accurately cites to the sources throughout the text.	One reference or citation is missing or incorrectly present.	Appropriate citing and reference list.	References and citations are missing or errors detract significantly from paper.	Part is missing or plagiarized. Incorrect citing.

Component Title	ASSIGNMENT				Percentage (%)	100
Criteria	Score and Descriptors					Weight (%) Marks
	Excellent (5)	Good (4)	Average (3)	Need Improvement (2)	Poor (1)	
Abstract						10
Introduction & Theory						10
Methodology/ Experimental						35
Results, Disscussion Conclusion						40
Formatting References						5
					TOTAL	100